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TECH HUB

• ELECTRICAL ENGINEERING • LESSON 03

# Ohm's Law — Series & Parallel

V=IR · KVL · KCL · Voltage Divider · Circuit Solving

Lesson 03 of 30 | Electrical Engineering Program

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- Series circuits — same current, voltages add (KVL)
- Parallel circuits — same voltage, currents add (KCL)
- KVL and KCL diagrams with worked verification
- 6-step systematic method for solving any DC circuit
- 4 practical examples: cable sizing, fuse selection, voltage drop

## Series Circuits — One Path for Current

In a **series circuit**, all components are connected end-to-end in a single loop. There is only one path for current to flow. This leads to two fundamental rules: the **current is identical everywhere** in the loop, and the **voltages across each component add up to the supply voltage**.

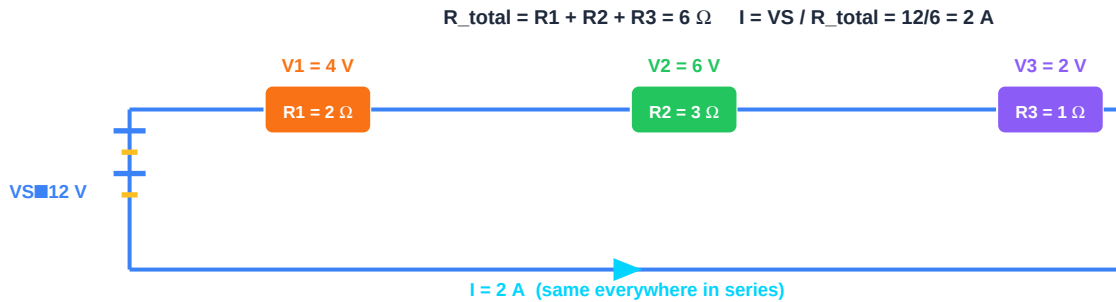


Figure 1 — Series Circuit: Same Current, Voltages Add (KVL:  $4+6+2 = 12\text{ V}$ )

| Series Circuit Rule               | Formula                                  | Worked Example ( $V_S=12\text{V}$ , $R_1=2\Omega$ , $R_2=3\Omega$ , $R_3=1\Omega$ )  |
|-----------------------------------|------------------------------------------|--------------------------------------------------------------------------------------|
| Total resistance                  | $R_T = R_1 + R_2 + R_3 + \dots$          | $R_T = 2 + 3 + 1 = 6\ \Omega$                                                        |
| Circuit current (same everywhere) | $I = V_S / R_T$                          | $I = 12 / 6 = 2\text{ A}$                                                            |
| Voltage across each resistor      | $V_n = I \times R_n$                     | $V_1=2 \times 2=4\text{V}$ , $V_2=2 \times 3=6\text{V}$ , $V_3=2 \times 1=2\text{V}$ |
| Voltages sum to supply (KVL)      | $V_S = V_1 + V_2 + V_3$                  | $12 = 4 + 6 + 2\ \checkmark$                                                         |
| Power in each resistor            | $P_n = I^2 \times R_n$ or $V_n \times I$ | $P_1=4\text{W}$ , $P_2=12\text{W}$ , $P_3=4\text{W}$ , Total=20W                     |
| Total power check                 | $P_T = V_S \times I$                     | $P_T = 12 \times 2 = 24\text{ W}$ (=4+12+4+4W fuse etc.)                             |

### Voltage Divider Rule

The voltage divider is one of the most used circuit techniques. If two resistors  $R_1$  and  $R_2$  are in series across a supply  $V_S$ , the voltage across  $R_2$  is:  **$V_2 = V_S \times R_2 / (R_1 + R_2)$** . This is used in sensor circuits, reference voltage generation, and signal level adjustment.

| Application                 | $R_1$         | $R_2$         | $V_S$ | $V_{out} = V_S \times R_2 / (R_1 + R_2)$ |
|-----------------------------|---------------|---------------|-------|------------------------------------------|
| Halve a 5V signal           | 10 k $\Omega$ | 10 k $\Omega$ | 5 V   | $5 \times 10 / (10+10) = 2.5\text{ V}$   |
| Set threshold at 2/3 supply | 1 k $\Omega$  | 2 k $\Omega$  | 9 V   | $9 \times 2 / (1+2) = 6\text{ V}$        |
| 10:1 voltage divider        | 9 k $\Omega$  | 1 k $\Omega$  | 24 V  | $24 \times 1 / (9+1) = 2.4\text{ V}$     |

**Practical use: Christmas lights wired in series — one bulb fails (open circuit) and the whole string goes out. That is the downside of series circuits in power applications. For this reason, building lighting is always wired in parallel.**

## Parallel Circuits — Multiple Current Paths

In a **parallel circuit**, components are connected between the same two nodes — all sharing the same voltage. Current splits between branches, with more current taking the lower-resistance path. This is how all domestic wiring works: every appliance sees the full mains voltage, and adding more appliances increases total current without affecting others.

$1/R_{total} = 1/6 + 1/4 + 1/12 = 0.5 \Omega^{-1} \rightarrow R_{total} = 2 \Omega \quad I_T = 12/2 = 6 \text{ A}$

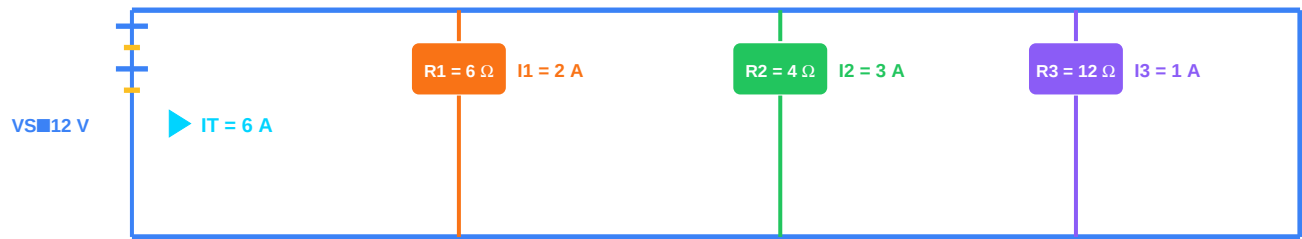


Figure 2 — Parallel Circuit: Same Voltage, Currents Add (KCL: 2+3+1 = 6 A)

| Parallel Circuit Rule           | Formula                     | Worked Example (VS=12V, R1=6Ω, R2=4Ω, R3=12Ω) |
|---------------------------------|-----------------------------|-----------------------------------------------|
| Voltage across each branch      | V1 = V2 = V3 = VS           | All branches see 12 V                         |
| Current in each branch          | I_n = VS / R_n              | I1=12/6=2A, I2=12/4=3A, I3=12/12=1A           |
| Total current (KCL)             | IT = I1 + I2 + I3           | IT = 2 + 3 + 1 = 6 A                          |
| Total (equivalent) resistance   | 1/R_T = 1/R1 + 1/R2 + 1/R3  | 1/R_T = 1/6+1/4+1/12 = 0.5 → R_T = 2 Ω        |
| R_T always less than smallest R | R_T < min(R1, R2, R3)       | R_T = 2 Ω < R1=6 Ω ✓                          |
| For 2 resistors only            | R_T = (R1 × R2) / (R1 + R2) | Product-over-sum shortcut                     |

### Current Divider Rule

For two parallel resistors R1 and R2 carrying total current IT: **I1 = IT × R2 / (R1 + R2)** and **I2 = IT × R1 / (R1 + R2)**. Note: current through R1 depends on R2 — higher R2 forces more current through R1 (the lower path).

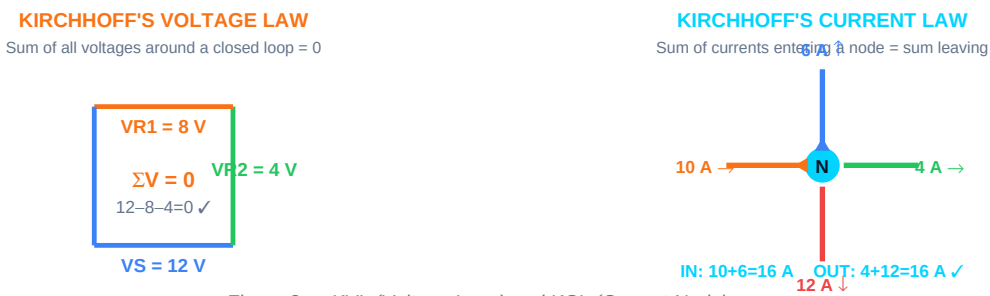


Figure 3 — KVL (Voltage Loop) and KCL (Current Node)

**KVL:** Going around any closed loop and summing voltage rises (+) and drops (–) always equals zero. **KCL:** At any node, the sum of currents entering equals the sum of currents leaving. These two laws, combined with Ohm's Law, let you solve any DC circuit no matter how complex.

## Solving Any DC Circuit — Systematic Method

With Ohm's Law + KVL + KCL you can solve any circuit. The key is a **disciplined step-by-step approach**: label unknowns, write the equations, solve algebraically. Rushing straight to numbers is the most common exam mistake.

| Step | Action                           | Detail                                                                                                                                                              |
|------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Draw and label the circuit       | Mark all components, nodes (letters A, B, C), and assumed current directions. If a current turns out negative, it flows opposite to your assumption — that is fine. |
| 2    | Identify unknowns                | List what you need to find: unknown currents (I1, I2...) and unknown voltages (V1, V2...). Count them — you need exactly that many independent equations.           |
| 3    | Apply KCL at nodes               | At each node: currents in = currents out. For N nodes, write N–1 independent KCL equations.                                                                         |
| 4    | Apply KVL around loops           | For each independent loop: $\Sigma V = 0$ . Voltage rise = + (going from – to + through source); Voltage drop = – (going through resistor in direction of current). |
| 5    | Apply Ohm's Law                  | Substitute $V = IR$ to convert voltage unknowns to current unknowns (or vice versa). Now all equations share the same unknowns.                                     |
| 6    | Solve the simultaneous equations | Use substitution, elimination, or matrix methods. Verify: check that KCL and KVL are satisfied with your answers.                                                   |

### Practical Applications of Ohm's Law

| Application                                                              | Calculation                                                                                                                                                                                                                                                          | Result                                                                                               |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Size a cable for a 3 kW heater at 230 V (max 2% voltage drop, 20 m run)  | $I = P/V = 3000/230 = 13.04\text{ A}$ Max VD = $0.02 \times 230 = 4.6\text{ V}$ Max R = $VD/I = 4.6/13.04 = 0.353\ \Omega$<br>Round trip: $2 \times 20 = 40\text{ m}$ Required $\rho L/A \leq 0.353\ \Omega$<br>$A \geq \rho L/R = (1.72\text{e-}8 \times 40)/0.353$ | Min cable CSA = $1.95\text{ mm}^2$<br>Use: $2.5\text{ mm}^2$ copper (standard next size up)          |
| Select a fuse for a 500 W, 230 V motor (starting current = 6× FLC)       | FLC = $P/V = 500/230 = 2.17\text{ A}$ Start current = $6 \times 2.17 = 13\text{ A}$ Fuse must carry FLC but blow on fault                                                                                                                                            | Use: 4 A slow-blow fuse (carries start surge; blows on short circuit)                                |
| Find voltage at end of a 50 m $2.5\text{ mm}^2$ cable, $I = 10\text{ A}$ | $R = \rho L/A = (1.72\text{e-}8 \times 100) / 2.5\text{e-}6 = 0.688\ \Omega$ (round trip $2 \times 50\text{m}$ ) VD = $I \times R = 10 \times 0.688 = 6.88\text{ V}$                                                                                                 | $V_{\text{load}} = 230 - 6.88 = 223\text{ V}$ (3.0% drop — at limit; consider $4\text{ mm}^2$ cable) |
| Parallel lamp circuit: 4 × 60 W lamps at 230 V — total current?          | Each lamp: $I = P/V = 60/230 = 0.261\text{ A}$ Parallel: $I_T = 4 \times 0.261$                                                                                                                                                                                      | $I_T = 1.04\text{ A}$ Total power = $240\text{ W}$ Fuse: 2 A (next standard size)                    |

### Series vs Parallel — Key Differences Summary

| Property        | Series                                     | Parallel                                      |
|-----------------|--------------------------------------------|-----------------------------------------------|
| Current         | Same through all components                | Splits — different in each branch             |
| Voltage         | Divides — adds up to VS                    | Same across all branches = VS                 |
| Total R         | $R_T = R_1 + R_2 + R_3$ (always increases) | $R_T < \text{smallest } R$ (always decreases) |
| One open fault  | All components off                         | Only that branch fails; others continue       |
| One short fault | All voltage appears across others          | That branch draws max current; fuse blows     |
| Typical use     | Fuses, indicator lamps (control)           | Power circuits, lighting, socket circuits     |

*What comes next — ELEC-004: AC vs DC — why AC is used for power distribution, transformers, single-phase and three-phase systems, power factor, and reactive power explained.*